

AssetFlow - Django Architecture Documentation

Project Models

accounts, dashboard, organization, assets, allocation, booking, maintenance, audit, reports, notifications, activitylogs

Roles

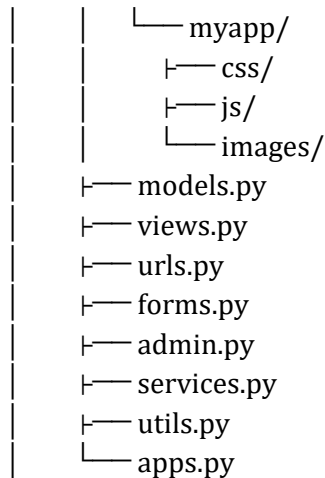
Admin, Asset Manager, Department Head, Employee

Core Models

User, Employee, Department, AssetCategory, Asset, AssetAllocation, TransferRequest, Resource, Booking, MaintenanceRequest, AuditCycle, AuditAsset, Notification, ActivityLog

Folder Structure

```
hack_env/
|
├── myproject/
│   ├── manage.py
│   ├── myproject/
│   │   ├── settings.py
│   │   ├── urls.py
│   │   ├── asgi.py
│   │   └── wsgi.py
│   └── myapp/
│       ├── migrations
│       ├── templates/
│       │   ├── myapp/
│       │   │   ├── login.html
│       │   │   ├── dashboard.html
│       │   │   ├── organization/
│       │   │   ├── assets/
│       │   │   ├── allocation/
│       │   │   ├── booking/
│       │   │   ├── maintenance/
│       │   │   ├── audit/
│       │   │   ├── reports/
│       │   └── notifications/
│       └── static/
```



Workflow

1. Employee Signup/Login
2. Admin creates Departments, Categories and promotes Asset Managers/Department Heads.
3. Asset Manager registers assets (QR, Asset Tag, Documents).
4. Allocate assets to employees/departments (double allocation blocked).
5. Employees book shared resources (overlap validation).
6. Maintenance Request -> Approval -> Technician -> Resolved.
7. Transfer Request -> Approval -> Asset Reallocated.
8. Return Asset -> Condition Check -> Available.
9. Audit Cycle -> Verify -> Discrepancy -> Close Audit.
10. Dashboard, Reports, Notifications and Activity Logs updated automatically.
11. AI Modules: Asset Health Score, Predictive Maintenance, Smart Dashboard, AI Chatbot, Suspicious Activity Detection.

Modules

Accounts, Dashboard, Organization Setup, Asset Management, Allocation, Booking, Maintenance, Audit, Reports, Notifications, Activity Logs, AI Engine.